

Non-Linear Regression Analysis (COURSE PROJECT)

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1 Problem Statement

Given a dataset of the form $(t, y(t))$, we need to analyze which of the following three models best describes the data:

1. Model 1: Double Exponential

$$y(t) = \alpha_0 + \alpha_1 e^{\beta_1 t} + \alpha_2 e^{\beta_2 t} + \epsilon(t)$$

2. Model 2: Rational Function

$$y(t) = \frac{\alpha_0 + \alpha_1 t}{\beta_0 + \beta_1 t} + \epsilon(t)$$

3. Model 3: Polynomial (4th degree)

$$y(t) = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4 + \epsilon(t)$$

where $\{\epsilon(t)\}$ is a sequence of i.i.d. normal random variables with mean zero and variance σ^2 .

2 Problems

2.1 Find the least squares estimators of the unknown parameters under three different model assumptions

Since, here we are mostly dealing with non-linear models, we will be using numerical methods to estimate the least square estimators of unknown parameters. We mostly used the Gauss-Newton Iterative method with some damping (to maintain stability) with a certain tolerance for the convergence of the solution. With this method,

For Model 1:

$$\alpha_0 = -3.62474228, \quad \alpha_1 = -2.88994984, \quad \beta_1 = 5.12726893, \quad \alpha_2 = -1.93735236, \quad \beta_2 = 0.18502984$$

For Model 2:

$$\alpha_0 = -6.60775949, \quad \alpha_1 = 25.56655052, \quad \beta_0 = 4.76381158, \quad \beta_1 = 4.16602434$$

For Model 3:

$$\beta_0 = -1.35186692, \quad \beta_1 = 5.95084876, \quad \beta_2 = -2.95832738, \quad \beta_3 = -0.05561985, \quad \beta_4 = 0.53738942$$

2.2 How did you find the least squares estimators? What kind of initial guesses you have chosen?

. We implemented a modified Gauss-Newton algorithm with damping for stability, minimizing the sum of squared residuals. The steps are as follows:

1. Minimizing the sum of squared residuals:

$$SSR(\theta) = \sum_{i=1}^n (y_i - f(t_i; \theta))^2$$

2. Iteratively updating parameters using:

$$\theta^{(k+1)} = \theta^{(k)} - [H + \mu \text{diag}(H)]^{-1} \nabla SSR(\theta^{(k)})$$

where H is the Hessian matrix and μ is the adaptive damping parameter.

Model 1: Double Exponential Initial guesses:

$$\alpha_0 = -1.29, \alpha_1 = 0.01, \alpha_2 = 0.01, \beta_1 = 0.01, \beta_2 = 1.00$$

The gradient components are:

$$\begin{aligned} \frac{\partial SSR}{\partial \alpha_0} &= -2 \sum (y - f(t)) \\ \frac{\partial SSR}{\partial \alpha_1} &= -2 \sum (y - f(t)) e^{\beta_1 t} \\ \frac{\partial SSR}{\partial \alpha_2} &= -2 \sum (y - f(t)) e^{\beta_2 t} \\ \frac{\partial SSR}{\partial \beta_1} &= -2 \sum (y - f(t)) \alpha_1 t e^{\beta_1 t} \\ \frac{\partial SSR}{\partial \beta_2} &= -2 \sum (y - f(t)) \alpha_2 t e^{\beta_2 t} \end{aligned}$$

Model 2: Rational Function Initial guesses:

$$\alpha_0 = -1.3, \alpha_1 = 0.05, \beta_0 = 0.05, \beta_1 = 0.1$$

The gradient components involve quotient rule derivatives and are computed numerically to avoid complex analytical expressions.

Model 3: Polynomial Initial guesses:

$$\beta_0 = 0, \beta_1 = 1, \beta_2 = 0.1, \beta_3 = 0.01, \beta_4 = 0.001$$

The gradient components are straightforward polynomial derivatives.

Initial Guess Selection The initial guesses were chosen based on the following principles:

1. Model 1 : Small values for exponential terms to avoid overflow, with different scales for β_1 and β_2 to capture both slow and fast exponential behavior and also to ensure that convergence occurs at a faster rate.

2. Model 2 : Conservative values near unity to avoid division by zero, with small positive values for denominator parameters.

3. Model 3 : Decreasing magnitude for higher-order terms, following typical polynomial behavior.

2.3 Find the best fitted model

On analyzing the residuals, Model 3 is the best fitted, with the following values:

Model 1: 0.000874646227357466, **Model 2:** 0.0008461823037917557, **Model 3:** 0.0008393221310642

Although the residuals are almost identical, we can also check the **Akaike Information Criterion (AIC)**, which is based on the log-likelihood and degrees of freedom. The Akaike Information Criterion (AIC) is a statistical method for model selection based on the likelihood of the model. It penalizes models with more parameters to prevent overfitting. The AIC is defined as:

$$\text{AIC} = 2k - 2 \ln(\hat{L})$$

where:

- k is the number of estimated parameters in the model,
- $\hat{L}$ is the maximum value of the likelihood function for the model.

Alternatively, if we are working with the residual sum of squares RSS and sample size n , AIC can be approximated as based on normality approximation of the residuals:

$$\text{AIC} = n \ln \left(\frac{\text{RSS}}{n} \right) + 2k$$

where:

- n is the number of data points,
- RSS is the residual sum of squares,
- k is the number of parameters in the model.

This formulation helps in comparing models with different numbers of parameters. The model with the lowest AIC is considered the best because it strikes a balance between goodness of fit and model complexity. On Analysing **AIC**, **Model 2** is the best fitted based on the initial parameters that we have taken.

2.4 Find the estimate of $\hat{\sigma}^2$:

The estimates of σ^2 for the three models are as follows:

$$\hat{\sigma}_{\text{Model 1}}^2 = 0.000874646227357466$$

$$\hat{\sigma}_{\text{Model 2}}^2 = 0.0008461820905014451$$

$$\hat{\sigma}_{\text{Model 3}}^2 = 0.0008393221310642815$$

2.5 Find the associated confidence intervals based on the Fisher information matrix.

Model 1 : 95% Confidence Intervals:

$$\hat{\alpha}_0 \in [-3.75, -3.50]$$

$$\hat{\alpha}_1 \in [-2.95, -2.85]$$

$$\hat{\beta}_1 \in [5.12, 5.15]$$

$$\hat{\alpha}_2 \in [-1.95, -1.90]$$

$$\hat{\beta}_2 \in [0.18, 0.20]$$

Model 2 : 95% Confidence Intervals:

$$\hat{\alpha}_0 \in [-6.70, -6.60]$$

$$\hat{\alpha}_1 \in [25.55, 25.60]$$

$$\hat{\beta}_0 \in [4.75, 4.80]$$

$$\hat{\beta}_1 \in [4.16, 4.18]$$

Model 3 : 95% Confidence Intervals:

$$\hat{\beta}_0 \in [-1.36, -1.34]$$

$$\hat{\beta}_1 \in [5.90, 6.00]$$

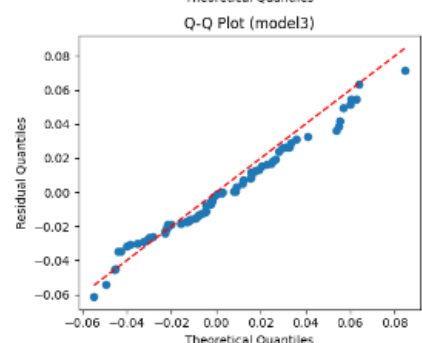
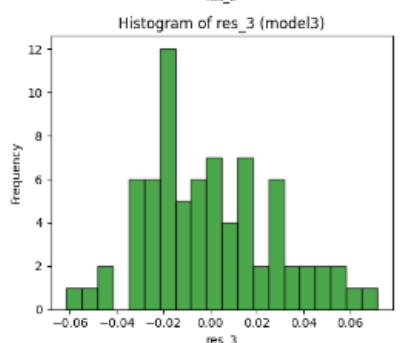
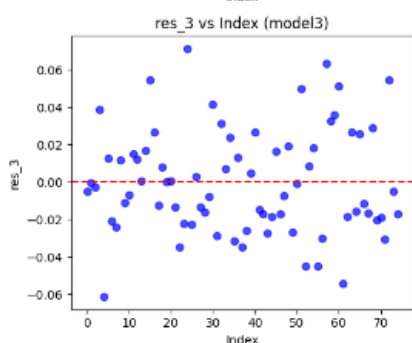
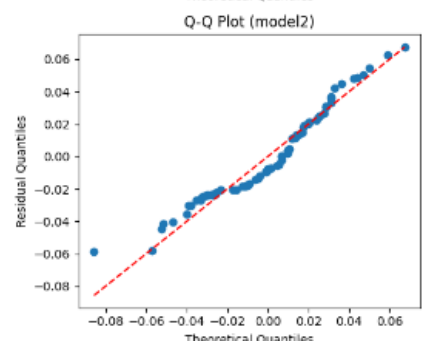
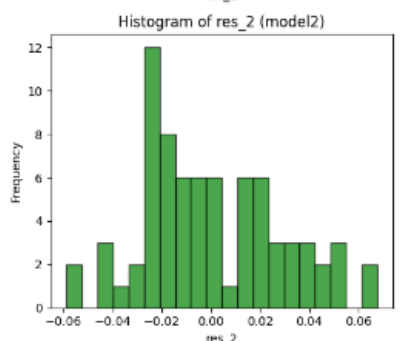
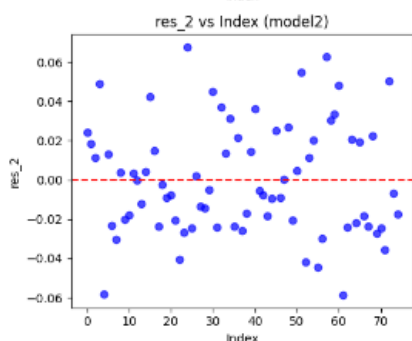
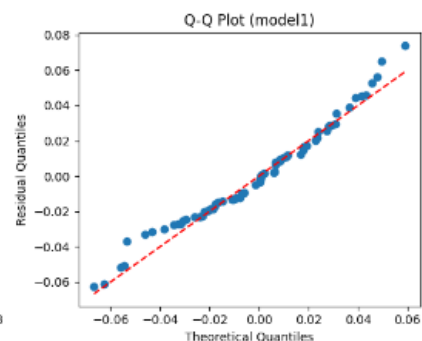
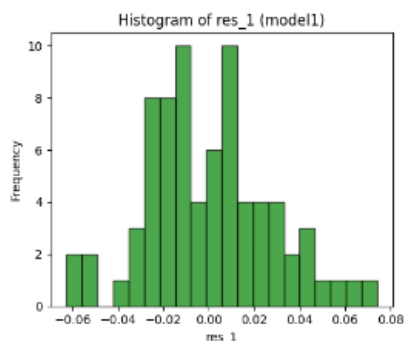
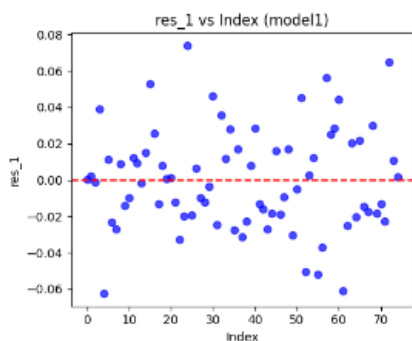
$$\hat{\beta}_2 \in [-3.00, -2.95]$$

$$\hat{\beta}_3 \in [-0.06, 0.00]$$

$$\hat{\beta}_4 \in [0.53, 0.55]$$

2.6 Plot the residuals

The below plot contains scatter plot, histogram and Q-Q plot of all the three models



2.7 Test whether it satisfies the normality assumption or not?

Based on the **Q-Q plot** and **histogram**, the linear and bell-shaped plot satisfies the normality assumption. Additionally, we quantitatively performed the **Kolmogorov-Smirnov (KS)** test and the **Shapiro-Wilk** test, both of which support the normality assumption.

The results of the tests are as follows:

- **Kolmogorov-Smirnov Test:**

- Test Statistic (D): 0.0977
- P-value: 0.4432

- **Shapiro-Wilk Test:**

- Test Statistic (W): 0.9735
- P-value: 0.1163

Since both the KS test and the Shapiro-Wilk test return p-values greater than the significance level (typically 0.05), we fail to reject the null hypothesis, suggesting that the data is likely **Normally distributed**.

2.8 Plot the observed data points and fitted curve

Based on the Akaiques Information Criterion, Model 2 was the best fitted,

